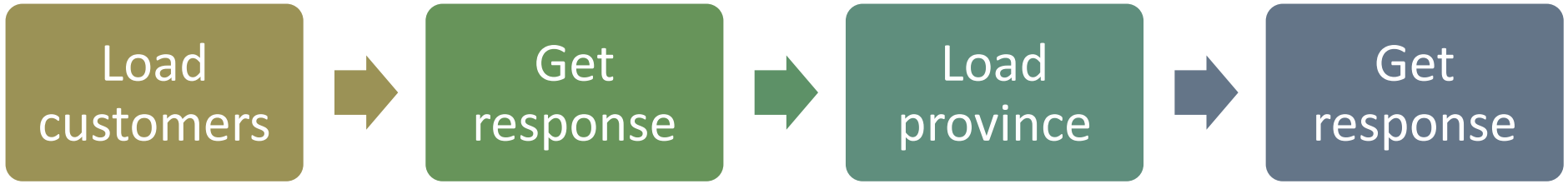

Frontend

Fetch data from api

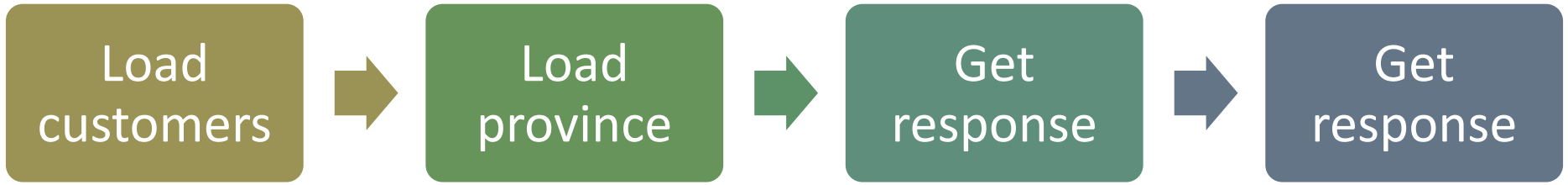
```
  // load customerList provinceList
  useEffect(() => {
    const fetchData = async () => {
      try {
        const customerResponse = await
axios.get("http://localhost:5000/api/customers");
                              setCustomerList(customerResponse.data);
        const provinceResponse = await
axios.get("http://localhost:5000/api/provinces");
                             setProvinceList(provinceResponse.data);
      } catch (error) {
        console.error("Error fetching data:", error);
      }
    };
    fetchData(); // Call the async function
```

await

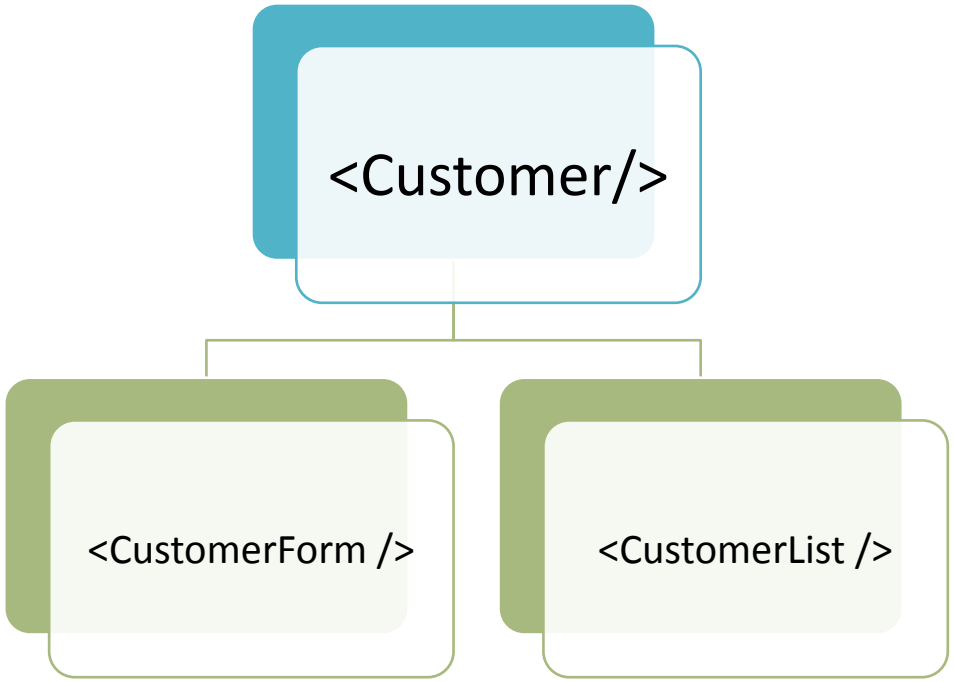


Promise.all

```
// load customerList provinceList
useEffect(() => {
  const getCustomerList = async () => {
    try {
      const [customerResponse, provinceResponse] = await Promise.all([
        axios.get("http://localhost:5000/api/customers"),
        axios.get("http://localhost:5000/api/provinces")
      ]);
                                setCustomerList(customerResponse.data);
                                setProvinceList(provinceResponse.data);
    } catch (error) {
      console.error("Error fetching data:", error);
    }
  };
  getCustomerList(); // Call the async function
}, []);
```



React



CustomerContainer

First Name

Last Name

Province กรุงเทพ

Add new

CustomerList

CustomerForm

Our Customer

- id = 1name = วิรัช ทองมาprovince = 1 del
- id = 2name = วิภา ทองใสprovince = 3 del

Customer

```
import Axios from 'axios'
import { useState, useEffect } from 'react’

import CustomerList from './CustomerList.jsx'
import CustomerForm from './CustomerForm.jsx’

function Customer() {
  const [customerList, setCustomerList] = useState([]);
  const [provinceList, setProvinceList] = useState([]);
  // fetchCustomers function

     // fetchProvinces function

  // load customerList provinceList

  // delete customer

  // add customer

  return (<>
      <CustomerForm addCustomer={addCustomer} provinceList={provinceList} />
      <CustomerList customerList={customerList} deleteCustomer={deleteCustomer} />
```

```
// fetchCustomers
  const fetchCustomers = async () => {
    try {
      const response = await axios.get("http://localhost:5000/api/customers");
      setCustomerList(response.data);
    } catch (error) {
      console.error("Failed to fetch customers:", error);
    }
  };
```

```
  // fetchProvinces
  const fetchProvinces = async () => {
    try {
      const response = await
axios.get("http://localhost:5000/api/provinces");
      setProvinceList(response.data);
    } catch (error) {
      console.error("Failed to fetch provinces:", error);
    }
  };
```

```
// fetch customers and provinces on component mount
useEffect(() => {
  const fetchData = async () => {
    await fetchCustomers();
    await fetchProvinces();
  };
  fetchData();
}, []);
```

```
// delete customer
const deleteCustomer = async (id) => {
  try {
    await axios.delete(`http://localhost:5000/api/customers/${id}`);
    setCustomerList((customerList) =>
      customerList.filter((customer) => customer.id !== id),
    );
  } catch (error) {
    console.error("Failed to delete customer:", error);
  }
};
```

```
// add customer
const addCustomer = async (newCustomer) => {
  try {
    await axios.post("http://localhost:5000/api/customers", newCustomer);
    await fetchCustomers();
  } catch (error) {
    console.error("Failed to add customer:", error);
  }
};
```

CustomerList

```
const CustomerList = ({ customerList, deleteCustomer }) => {
    return (<div>
        <h5>Our Customer</h5>
        <ul>
            {customerList.map(
                (item, index) => (<li key={index}>
                    id = {item.customerId}
                    name = {item.firstName} {item.lastName}

                    province = {item.provinces == null ? " - " : item.provinces.name}

                    <button onClick={() => deleteCustomer(item.customerId)}>
                  delete</button></li>)
            )}
        </ul>
    </div>
    );
}
export default CustomerList;
```

Our Customer

- id = 1name = วิรัช ทองมาprovince = กรุงเทพ del
- id = 2name = วิภา ทองใสprovince = นนทบุรี del

CustomerForm

```
import { useState } from "react";
const CustomerForm = ({ addCustomer, provinceList }) => {

  // state for form inputs


  // handle input changes


  // handle form submission


    // render the form


 };
export default CustomerForm;
```

```
// state for form inputs
const [inputs, setInputs] = useState({
  id: "",
  firstName: "",
  lastName: "",
  provinceId: "",
});
```

```
// handle input changes
function handleChange(e) {
  const { name, value } = e.target;
  setInputs((prevInputs) => ({
    ...prevInputs,
    [name]: value,
  }));
}
```

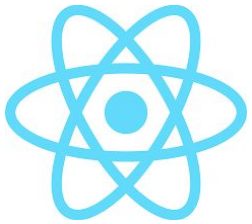
```
// handle form submission
const handleSubmit = async (event) => {
  event.preventDefault();

  await addCustomer({
    fName: inputs.firstName,
    lName: inputs.lastName,
    province: inputs.provinceId,
  });

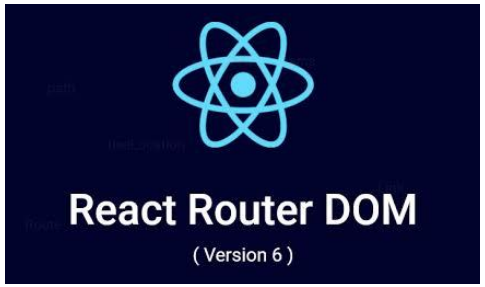

  setInputs({ id: "", firstName: "", lastName: "", provinceId: "" });
};
```

```
  // render the form

  return (
    <>
      <h2>customer form</h2>
      <form onSubmit={handleSubmit}>
          <label> First Name
            <input type="text" name="firstName" value={inputs.firstName}
onChange={handleChange} />
          </label>
          <label> Last Name
            <input type="text" name="lastName" value={inputs.lastName}
onChange={handleChange} />
          </label>
          <label> Province
            <select name="provinceId" value={inputs.provinceId} onChange={handleChange} >
              {provinceList.map((item, index) => (
                <option value={item.provinceId}>{item.name}</option>
              ))}
            </select>
          </label>
        <button type="submit">Add new</button>
      </form>
    </>
  );
```



AXIOS

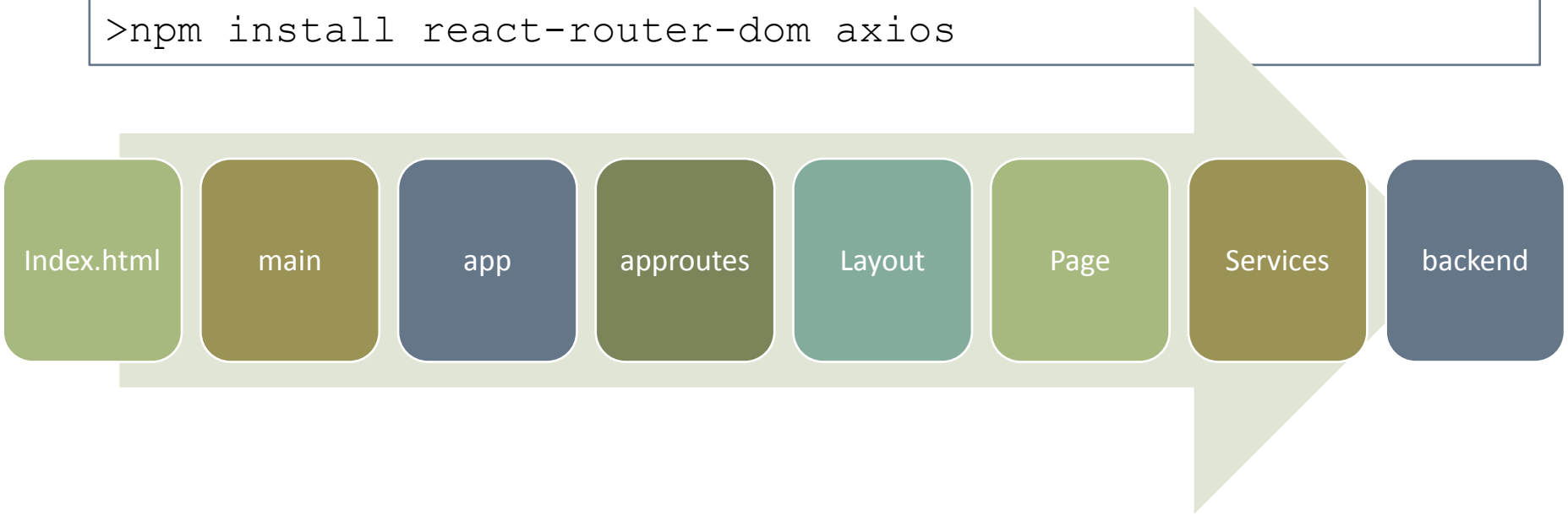


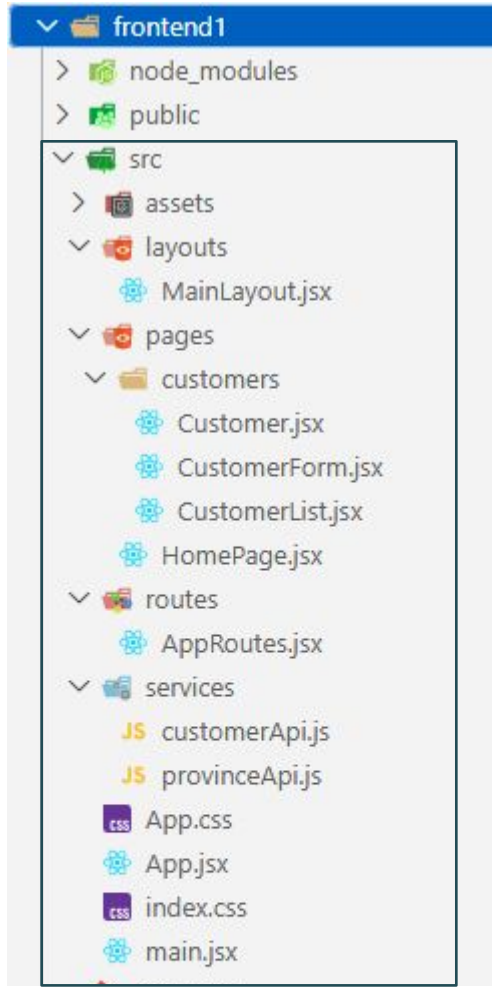

```
// Install ReactJs inside frontend

>npm install -g create-vite

>npm create vite@latest


>npm install react-router-dom axios
```





main.jsx

```jsx
import { createRoot } from 'react-dom/client'
import { StrictMode } from 'react'
import { BrowserRouter } from 'react-router-dom';
import App from './App.jsx'
import './index.css'


createRoot(document.getElementById('root')).render(
  <StrictMode>
    <BrowserRouter>
      <App />
    </BrowserRouter>
  </StrictMode>,
)
```

App.jsx

```jsx
import AppRoutes from "./routes/AppRoutes";


function App() {
  return (
    <AppRoutes />
  );
}


export default App;
```

/src/routes/AppRoutes.jsx

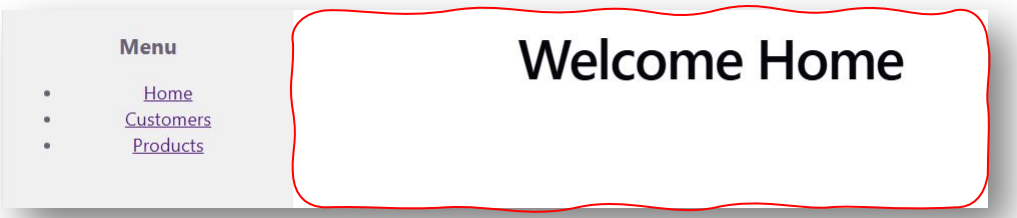
AppRoutes

```
import { Routes, Route } from "react-router-dom";
import MainLayout from "../layouts/MainLayout";
import HomePage from "../pages/HomePage";
import Customer from "../pages/customers/Customer";

function AppRoutes() {
  return (
    <Routes>
      <Route element={<MainLayout />}>
        <Route path="/" element={<HomePage />} />
        <Route path="/customers" element={<Customer
/>} />
      </Route>
    </Routes>
  );
}

export default AppRoutes;
```

MainLayout

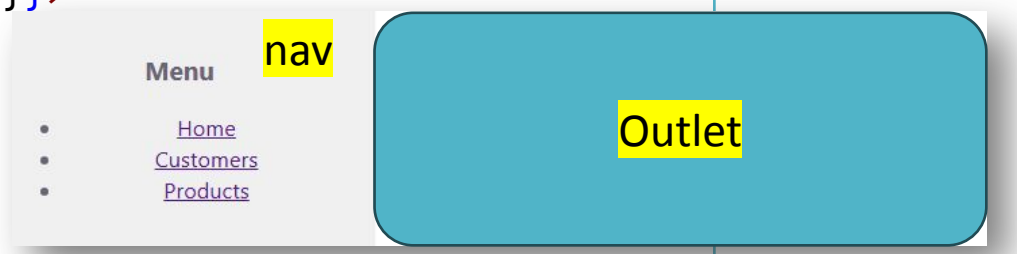


/src/layouts/MainLayout.js

x

Layout

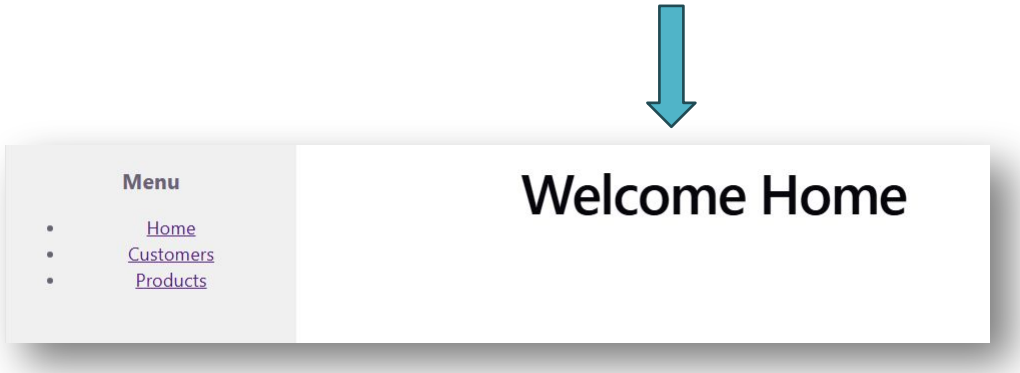
```
import { Outlet, Link } from "react-router-dom";
function MainLayout() {
  return (
    <div style={{ display: 'flex', height: '100vh' }}>
      <nav style={{ width: '250px', background: '#f0f0f0', padding: '20px'
}}>
        <h3>Menu</h3>
        <ul>
          <li><Link to="/">Home</Link></li>
          <li><Link to="/customers">Customers</Link></li>
          <li><Link to="/products">Products</Link></li>
        </ul>
      </nav>
      <main style={{ flex: 1, padding: '20px' }}>
        <Outlet />
      </main>
    </div>
  );
}
export default MainLayout;
```



src/pages/HomePage.jsx

HomePage

```jsx
export default function HomePage() {
  return (
    <div>
      <h1>Welcome Home</h1>
    </div>
  );
}
```

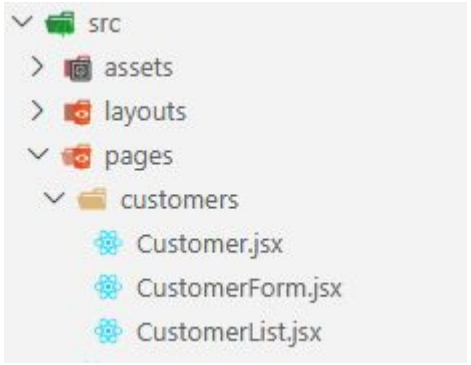


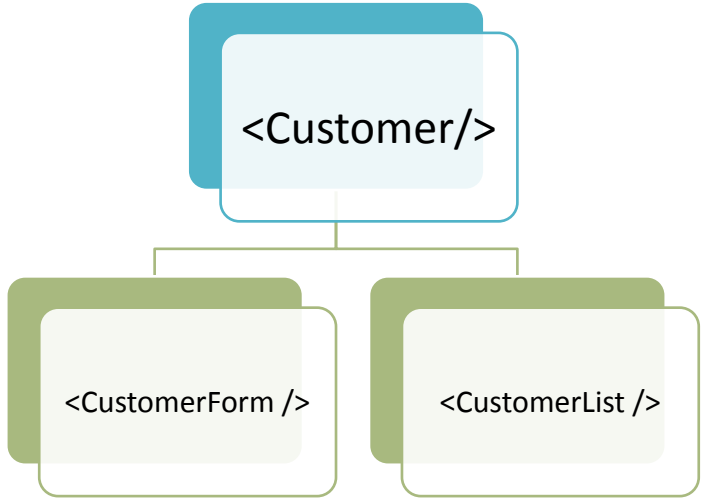
Services

/src/services/customerApi.j
s

```
import axios from 'axios';
const API_URL = 'http://localhost:5000/api/customers';
export const customerApi = {
  getAll: async () => {
    const response = await axios.get(API_URL);
    return response.data;
  },
  getById: async (id) => {
    const response = await axios.get(`${API_URL}/${id}`);
    return response.data;
  },
  create: async (data) => {
    const response = await axios.post(API_URL, data);
    return response.data;
  },
  update: async (id, data) => {
    const response = await axios.put(`${API_URL}/${id}`,
data);
    return response.data;
  },
  delete: async (id) => {
    const response = await axios.delete(`${API_URL}/${id}`);
    return response.data;
  },
};
```

Customer





customer.jsx

```
import { useState, useEffect } from "react";


import { customerApi } from
'../../services/customerApi';
import { provinceApi } from
'../../services/provinceApi';


import CustomerList from "./CustomerList";
import CustomerForm from "./CustomerForm";
```

fetchCustomers

```
try {
      const response = await axios.get("http://localhost:5000/api/customers");
      setCustomerList(response.data);
}
```



```
try {
      const data = await customerApi.getAll();
      setCustomerList(data)
}
```



```
getAll: async () => {
    const response = await
axios.get(API_URL);
    return response.data;
  }
```

fetchProvinces

```
    try {
      const data = await provinceApi.getAll();
      setProvinceList(data);
    }
```

deleteCustomer

```
await axios.delete(`http://localhost:5000/api/customers/${id}`);
```



```
await customerApi.delete(id);
```

addCustomer

```
await axios.post("http://localhost:5000/api/customers", newCustomer);
```



```
const data = await customerApi.create(newCustomer);
```

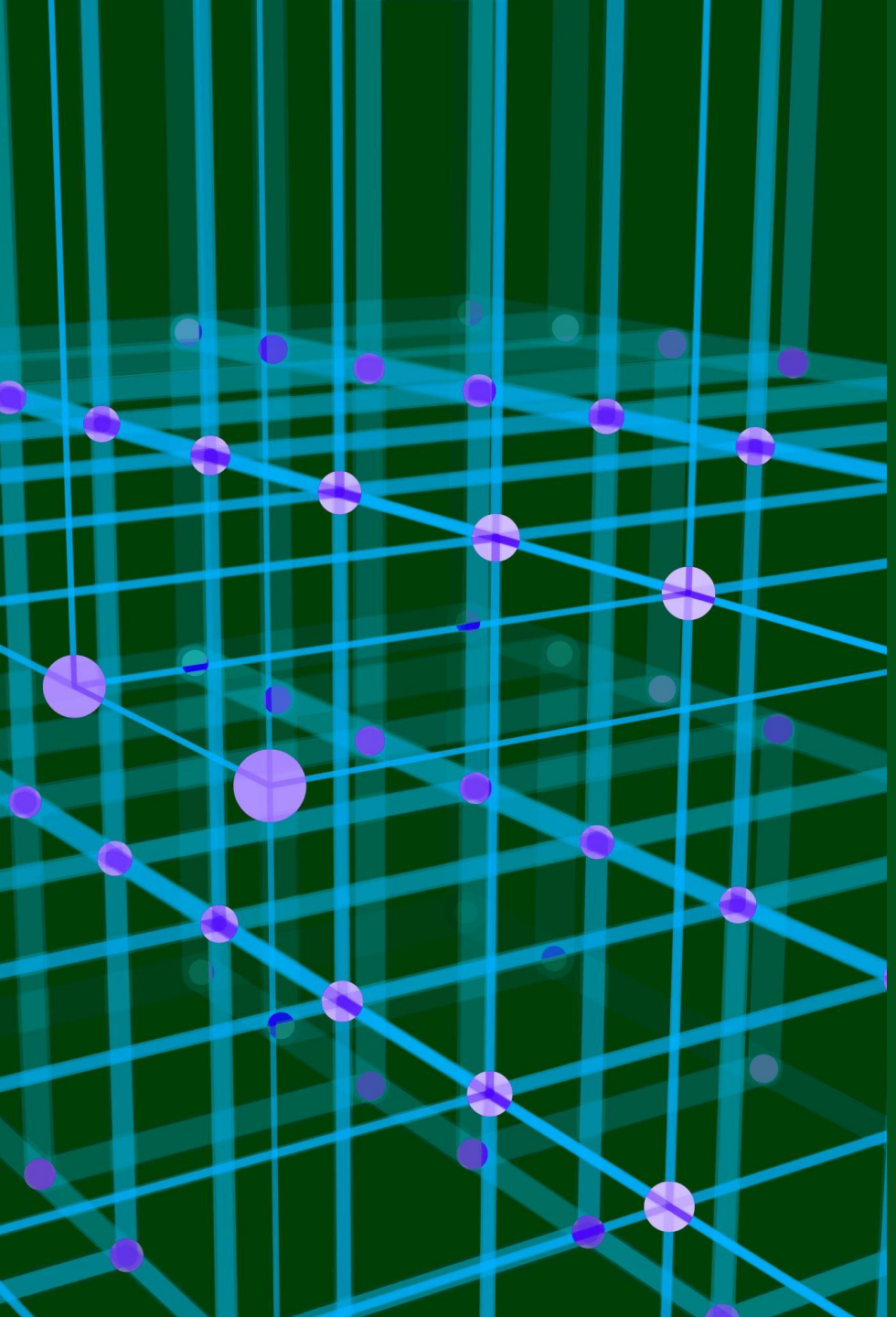
/src/routes/AppRoutes.jsx

Route not found

```
<Routes>
  <Route element={<MainLayout />}>
    <Route path="/" element={<HomePage />} />
    <Route path="/customers" element={<Customer />} />
  </Route>

      <Route
    path ="*"
    element={
      <div >
        <h1>404</h1>
        <p>ไม่พบหน้าที่คุณต้องการ</p>
        <Link to="/">กลับสู่หน้าหลัก</Link>
      </div>
    }
  />
</Routes>
```





Todo

ปรับแก้งานครั้งก่อน โดยให้เชื่อมต่อฐานข้อมูล ผ่าน backend เพื่อ เพิ่ม ลบ แก้ไขเมนูต่างๆ

